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Constructive approximation in mixed-norm spaces

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Abstract
The concept of mixed-norm spaces has emerged as a significant interest in fields such as harmonic analysis. In addition, the problem of function approximation through sampling series has been particularly noteworthy in approximation theory. In this paper, we study the approximation problem in diverse mixed-norm function spaces. We utilize the family of Kantorovich-type sampling operators as approximator for the functions in mixed-norm Lebesgue space $L^{\overrightarrow{P}}(\mathbb{R}^n)$, with $\overrightarrow{P} = (p_1, p_2, \ldots, p_n)$, and mixed-norm Orlicz space $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$, with $\overrightarrow{\Phi} = (\phi_1, \phi_2, \ldots, \phi_n)$, where each ϕ_i is an Orlicz function (defined in Section 2). The Orlicz spaces are a generalized family that encompasses many significant function spaces. We establish the boundedness of the family of generalized and Kantorovich-type sampling operators within the framework of these mixed-norm spaces. Further, we study the approximation properties of Kantorovich-type sampling operators in both mixed-norm Lebesgue and Orlicz spaces. Lastly, we provide several examples of appropriate kernels that demonstrate the applicability of the proposed theory.

Keywords Approximation of functions · Kantorovich-type sampling series · Mixed-norm Lebesgue space · Mixed-norm Orlicz space

Mathematics Subject Classification 94A20 · 26D15 · 46B09 · 46E30 · 47B34

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1 Introduction

In broad terms, the aim of function approximation is to identify a simpler function that closely represents a complicated function. Sampling and reconstruction of functions have been playing a pivotal role in approximation theory, serving as fundamental tools for accurately representing functions through discrete sample values [12]. The pioneer study of sampling and reconstruction problem for bandlimited functions was carried out in [9, 10, 49].

A fundamental challenge in signal processing is to recover a signal f from their discrete sample values. Whittaker-Kotelnikov-Shannon (WKS) [9, 16, 17] collectively provide a reconstruction formula to recover bandlimited signal from their discrete sample measurements. In particular, a bandlimited signal $f : \mathbb{R} \to \mathbb{C}$ with bandwidth $2\pi w$, for some $w > 0$, i.e., the Fourier transform of f is supported on $[-\pi w, \pi w]$, can be entirely reconstructed from their samples values at $(\frac{k}{w})_{k\in\mathbb{Z}}$ by using the formula

$$f(x) = \sum_{k\in\mathbb{Z}} f\left(\frac{k}{w}\right) \operatorname{sinc}(xw - k), \; x \in \mathbb{R}, \tag{1.1}$$

where $\operatorname{sinc}(x) := \frac{\sin \pi x}{\pi x}$ for $x \neq 0$ and $\operatorname{sinc}(0) = 1$. The WKS sampling theorem is the classical sampling theorem originally formulated by Shannon, serving as a fundamental concept in communication theory, information theory, and signal processing [9, 12, 33]. Moreover, the application is not limited to communication theory but also have remarkable advancements in areas such as optics, time-varying systems, boundary value problems, approximation and interpolation theory, and Fourier transform [12, 33, 49]. Some notable developments related to the WKS sampling theorem are detailed in [5, 9, 15, 17, 33].

Generalized sampling series. The WKS sampling theorem was generalized for the signals in the shift-invariant space, image of idempotent integral operator and so on [32]. The key concept is to recover the signal uniquely and stably from their sample values without losing any information. In (1.1), the sinc kernel is used despite its computational limitations and the fact that $\operatorname{sinc} t$ is not absolutely integrable over $\mathbb{R}$ [17]. To address this, Butzer [14, 16] introduced a generalized sampling series using the discrete sample values of the signal f and kernel χ. The generalized sampling series for a signal f is given by

$$S_w(f)(x) := \sum_{k\in\mathbb{Z}} f\left(\frac{k}{w}\right) \chi(wx - k), \quad x \in \mathbb{R}, \quad w > 0, \tag{1.2}$$

where χ denotes the kernel function satisfying certain assumptions [16]. The pointwise and uniform convergence for (1.2) was initially discussed in the foundational work of Butzer and Stens [14, 16]. Bardaro et al. [5] extended the classical Shannon sampling series (1.1) by introducing an L^p averaged modulus of smoothness to characterize intricate signals. The reconstruction of p-integrable functions by generalized sampling series was further investigated in [17]. Additionally, Butzer et al. [14]

studied the approximation properties of generalized sampling series for both continuous and discontinuous functions. In a subsequent work, the same author developed convergence results for the multivariate generalized sampling series in the space of continuous functions [11]. Further details on the properties and applications of these operator can be found in [5, 7, 10, 11, 14, 17].

Kantorovich-type sampling Series. Later, Bardaro et al. [6] introduced the Kantorovich-type sampling series. Instead of taking sample values, these series compute the average value of the function over each interval, namely $w\int_{\frac{k}{w}}^{\frac{k+1}{w}} f(u)du$, for $w>0$. For a locally integrable function $f:\mathbb{R}\to\mathbb{R}$, the Kantorovich-type sampling series is defined as

$$K_w(f)(x) := \sum_{k\in\mathbb{Z}} \chi(wx-k)\; w\int_{\frac{k}{w}}^{\frac{k+1}{w}} f(u)du, \quad x\in\mathbb{R}, \quad w>0. \tag{1.3}$$

The approximation of continuous functions by Kantorovich-type sampling series have been established in [6, 25], while the effectiveness of the series defined in (1.3) have also been demonstrated for reconstructing signals that may not be continuous [46]. Although the averaging techniques improve the regularity of these series, achieving convergence at the discontinuity points of f remains a challenge due to technical limitations. This issue is closely related to the application of Kantorovich-type sampling series in image reconstruction and enhancement.

Recently, Costarelli et al. [24] studied the approximation properties of (1.3) for discontinuous functions across various fields, including biomedical and engineering applications [18, 22]. Cantarini et al. [19] made a significant contribution to the study of both differentiable and not differentiable functions for (1.3). In [1], the convergence behavior of m- order Kantorovich-type sampling operators is discussed. The generalized Kantorovich-type sampling operators is introduced in [46], where local averages are determined through the convolution with an arbitrary function $\chi\in L^1(\mathbb{R})$. Bardaro et al. [6] pioneered the study of the convergence of the operators $(K_w)_{w>0}$ in the framework of Orlicz spaces. This line of research have been further extended to the multivariate form of Kantorovich-type sampling operators in [25]. In [20], the authors studied Kantorovich-type sampling operators in classical as well as fractional Sobolev-type spaces and established convergence theorems under appropriate kernel assumptions and introduced tight fractional Sobolev spaces for the analysis of the fractional case. Sharp strong and weak approximation results for Kantorovich-type neural network operators in Sobolev-Orlicz and Orlicz spaces are obtained in [26] along with convergence result, inverse theorems and characterizations of the corresponding Lipschitz classes. A characterization of generalized Lipschitz classes through the rate of convergence of Durrmeyer-type sampling operators in Sobolev spaces are established in [27]. The development of the theory in a multivariate framework is important for applications, particularly in signal theory and image processing. Several recent developments concerning the family (1.3) are discussed in [6, 19, 23–25, 46].

This paper aims to investigate the approximation properties of a family of sampling operators within the framework of a mixed-norm structure. We study its approximation capabilities in a broad class of functions spaces, which may not be continuous, such as mixed-norm Lebesgue spaces and mixed-norm Orlicz spaces. First, we examine the boundedness of generalized sampling operators in mixed-norm Lebesgue space. We then establish the boundedness and convergence of Kantorovich-type sampling operators within mixed-norm Lebesgue spaces. Further, we extend these results in the setting of mixed-norm Orlicz spaces.

The theory of *Orlicz spaces*, which provides a natural generalization of Lebesgue spaces, was established by W. Orlicz in [44]. It is noteworthy that the Orlicz spaces, denoted as L^{ϕ}, consist of many significant function spaces for different choice of $\phi-$functions, namely classical Lebesgue spaces, Exponential spaces and Logarithmic spaces (also known as Zygmund spaces). The seminal work on Orlicz spaces can be found in [37, 40], and since then, various researchers have studied their properties and potential applications in diverse areas. The rigorous framework of Orlicz spaces and their properties was introduced by Rao and Ren [47], along with related topics such as compact sets, the product of Orlicz spaces, topological structures, and more. Moreover, [48] provides a detailed discussion of fundamental applications of Orlicz spaces in areas such as Fourier analysis, stochastic analysis, and nonlinear PDE. Function spaces derived from Orlicz spaces have proven highly effective in the study of functional analysis and operator theory [28, 36]. Submultiplicative functions play an important role in interpolation theory and operator theory within Orlicz spaces, certain results are presented in [30]. Hence, Orlicz spaces offer a unified framework for studying Kantorovich-type sampling operators in different function spaces. In particular, mixed-norm function spaces are considered due to their flexible structure and broad applicability.

In *mixed-norm spaces*, the norm of measurable multivariable functions is determined by the iterative application of different norm functions. The foundation for this concept is found in the pioneering article by Benedek and Panzone [8]. For instance, unlike classical Lebesgue spaces, in mixed-norm Lebesgue spaces, the exponent for integrability can differ for each variable. This flexibility makes it more useful for areas where different variables may have different properties. These spaces are particularly relevant for time-varying signals [34] and playing a pivotal role in examining the solutions to partial differential equations that involve both time and spatial variables, such as the heat and wave equations [35]. In [3, 8, 21, 29, 31, 42], mixed-norm spaces are examined from an operator-theoretic perspective. The necessary and sufficient condition for the embedding of mixed-norm Lebesgue spaces and mixed-norm Orlicz spaces are discueed in [42]. Maligranda's work [41] focused on the Calderón–Lozanovskiĭ construction, particularly its application to mixed-norm function spaces generated from Banach ideal spaces. His research included a rigorous analysis of the Calderón product associated with these function spaces.

To the best of our knowledge, these operators have not been studied in mixed-norm Lebesgue spaces and mixed-norm Orlicz spaces. Given the enduring interest in Kantorovich-type sampling operators in approximation theory, examining their behavior in mixed-norm settings is of particular significance.

Contributions. The key features of the article are as follows.

- In [5, 6, 25], the boundedness as well as convergence of generalized and Kantorovich-type sampling series have been studied in Lebesgue space and Orlicz space. Here, we investigate these approximation properties for aforementioned family of sampling operators within mixed-norm structures. In particular, the mixed-norm Orlicz spaces have received limited attention in the literature, this study also contributes to the theoretical advancement of these spaces.
- This paper provides a broader framework for studying various classes of function spaces, due to the general structure of Orlicz spaces. On the way of establishing the convergence of Kantorovich-type sampling series in mixed-norm Orlicz space $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$, we prove a fundamental result asserting that, the space of compactly supported continuous functions $C_c(\mathbb{R}^n)$ is dense in mixed-norm Orlicz space $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$ (see Lemma 5.4).

Outline of the paper. The paper is structured as follows. In Section 2, we begin by recalling some fundamental definitions and results concerning mixed-norm Lebesgue spaces and Orlicz spaces, which are requisite for the proposed study. In Section 3, we prove the boundedness of generalized sampling operators in $L^{\overrightarrow{P}}(\mathbb{R}^n)$, followed by the boundedness and the convergence of Kantorovich-type sampling operators in $L^{\overrightarrow{P}}(\mathbb{R}^n)$ is proved in Section 4. In Section 5, we extend these results to the setting of mixed-norm Orlicz space $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$. In order to prove these results, we establish a density result in mixed-norm Orlicz space. In Section 6, we present several examples of kernel satisfying the assumptions (given in Section 2) of presented theory.

2 Preliminaries

In this section, we define some basic definitions and discuss preliminary results on mixed-norm Lebesgue space and mixed-norm Orlicz space.

2.1 Notations

The notations $\mathbb{N}^n$, $\mathbb{Z}^n$ and $\mathbb{R}^n$ represent the set of all ordered n-tuples of natural numbers, integers and real numbers respectively. Let $C(\mathbb{R}^n)$ denotes the set of all bounded and uniformly continuous functions, it also forms a normed vector space with the usual sup norm $\|f\|_\infty := \text{ess sup}_{\mathbf{t}\in\mathbb{R}^n} |f(\mathbf{t})|$. For $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$, we write $(\mathbf{x} + \mathbf{y}) = (x_1 + y_1, \ldots, x_n + y_n)$, and $\zeta\mathbf{x} = (\zeta x_1, \ldots, \zeta x_n)$, for any scalar $\zeta \in \mathbb{R}$. Here $\mathbf{x} = \mathbf{y}$ denotes that $x_i = y_i$ for each $i = 1, 2, \ldots, n$. The space $L^p(\mathbb{R}^n)$, for $1 \leq p < \infty$ contains of all p-integrable functions on $\mathbb{R}^n$ with the standard Lebesgue $p-$norm. The space $L^\infty(\mathbb{R}^n)$ refers to the collection of all measurable functions that are bounded and equipped with the sup norm $\|\cdot\|_\infty$. Here, we do not identify functions that are equal almost everywhere, since the sampling operator depends on pointwise evaluation of the function. Furthermore, $M(\mathbb{R}^n)$ is the space of all bounded and measurable real valued functions.

2.2 Mixed-norm Lebesgue space

Let (Ω_i, S_i, μ_i) be the σ-finite measure spaces, for $1 \leq i \leq n$, and (Ω, S, μ) represent corresponding product measure space, i.e., $\Omega = \prod_{i=1}^{n} \Omega_i$. The mixed-norm Lebesgue space is defined as follows.

Definition 2.1 (Mixed-norm Lebesgue space) For $1 \leq p_i \leq \infty$ and $\overrightarrow{P} = (p_1, \ldots, p_n)$, the mixed-norm Lebesgue space $L^{\overrightarrow{P}}(\Omega)$ is defined as

$$L^{\overrightarrow{P}}(\Omega) = \left\{ f : \Omega \to \mathbb{R} \text{ measurable } : \|f\|_{\overrightarrow{P}} < \infty \right\},$$

where

$$\|f\|_{\overrightarrow{P}} = \left(\int_{\Omega_n} \cdots \left(\int_{\Omega_2} \left(\int_{\Omega_1} |f(x_1, \ldots, x_n)|^{p_1} d\mu_1 \right)^{\frac{p_2}{p_1}} d\mu_2 \right)^{\frac{p_3}{p_2}} \ldots d\mu_n \right)^{\frac{1}{p_n}}.$$

The mixed-norm Lebesgue space can equivalently be viewed within the framework of vector-valued Lebesgue spaces, represented as follows:

$$L^{\overrightarrow{P}}(\Omega) = L^{p_n}_{x_n}\left(\Omega_n, L^{(p_1, p_2, \ldots, p_{n-1})}_{x_1, x_2, \ldots, x_{n-1}} \left(\prod_{k=1}^{n-1} \Omega_k \right) \right).$$

In the subsequent analysis, we only focus on the tuples $\overrightarrow{P} = (p_1, p_2, \ldots, p_n)$ with non-decreasing components, i.e., $p_i \leq p_{i+1}$ for $1 \leq i \leq n-1$. Although, this is a strong assumption and does not encompass all possible mixed-norm Lebesgue space $L^{\overrightarrow{P}}(\Omega)$, it still generalizes the classical Lebesgue space $(L^p(\Omega), \|\cdot\|_p)$. In particular, when $\overrightarrow{P} = (p, p, \ldots, p) \in [1, \infty)^n$ is constant vector, the mixed-norm space $\left(L^{\overrightarrow{P}}(\Omega), \|\cdot\|_{\overrightarrow{P}}\right)$ is equivalent to the standard Lebesgue space $(L^p(\Omega), \|\cdot\|_p)$. This monotonicity assumption on the exponents is not only analytically convenient but also appears in the study of sampling and reconstruction within reproducing kernel subspaces of mixed-norm Lebesgue space [38].

Definition 2.2 The space $\ell^{\overrightarrow{P}}(\mathbb{Z}^n)$ is a collection of sequences on $\mathbb{Z}^n$ with the mixed-norm defined as follows:

$$\|x\|_{\ell^{\overrightarrow{P}}} = \left(\sum_{k_n \in \mathbb{Z}} \cdots \left(\sum_{k_2 \in \mathbb{Z}} \left(\sum_{k_1 \in \mathbb{Z}} |x(k_1, k_2, \ldots, k_n)|^{p_1} \right)^{\frac{p_2}{p_1}} \right)^{\frac{p_3}{p_2}} \cdots \right)^{\frac{1}{p_n}} < \infty.$$

For a positive real w, we define the space of weighted sequence space $\ell^{\overrightarrow{P}}_w(\mathbb{Z}^n)$ such that

$$\|x\|_{\ell^{\overrightarrow{P}}_w} := \left(\frac{1}{w} \sum_{k_n \in \mathbb{Z}} \cdots \left(\frac{1}{w} \sum_{k_2 \in \mathbb{Z}} \left(\frac{1}{w} \sum_{k_1 \in \mathbb{Z}} |x(k_1, k_2, \ldots, k_n)|^{p_1} \right)^{\frac{p_2}{p_1}} \right)^{\frac{p_3}{p_2}} \cdots \right)^{\frac{1}{p_n}} < \infty.$$

For detailed information on mixed-norm Lebesgue spaces, we refer [2, 8, 21, 29, 41, 42, 50].

2.3 The Orlicz spaces

A convex and left-continuous function $\phi : [0, \infty) \to [0, \infty]$ is called an Orlicz function if it satisfies the following conditions:

(i) $\lim_{u\to\infty} \phi(u) = \infty$, and $\phi(0) = 0$,
(ii) $\phi(u) > 0$ for every $u > 0$, and is non-decreasing on $[0, \infty)$.

It is important to note that ϕ may take the value ∞ and is not necessarily continuous. For instance, for a fixed $b > 0$,

$$\phi(u) = \begin{cases} 0, & \text{if } u \leq b \\ \infty, & \text{if } u > b, \end{cases}$$

defines an Orlicz function. Orlicz functions generalize the power functions used in defining classical L^p spaces and allow for more flexible control over growth and integrability conditions.

To construct a function space associated with an Orlicz function ϕ, we introduce a fundamental building block: the modular functional associated with ϕ. Given a measurable space (Ω, μ), the functional $I^\phi : M(\Omega) \to [0, \infty]$ is defined by

$$I^\phi(f) := \int_\Omega \phi(|f(x)|) d\mu.$$

This functional generalizes the expression $\int |f|^p \, d\mu$ from classical Lebesgue theory and is central to defining the Orlicz space. The functional I^ϕ is called a modular functional and satisfies the following properties for every $f,\ g \in M(\Omega)$:

(i) $I^\phi(f) = 0$ if and only if $f = 0$,
(ii) $I^\phi(-f) = I^\phi(f)$,
(iii) $I^\phi(\gamma f + \beta g) \leq I^\phi(f) + I^\phi(g),\ \ for\ \gamma, \beta \geq 0,\ \ \gamma + \beta = 1.$

Moreover, for any $f \in M(\Omega)$, the map $\alpha \mapsto I^\phi(\alpha f)$ is non-decreasing for $\alpha > 0$.

Definition 2.3 (Orlicz space) Let ϕ be an Orlicz function. Then the Orlicz space generated by ϕ is given as

$$L^\phi(\Omega) = \left\{ f \in M(\Omega) : I^\phi(\lambda f) < \infty \ \text{ for some } \lambda > 0 \right\},$$

equipped with the norm

$$\|f\|_\phi = \inf \left\{ \lambda > 0 : \int_\Omega \phi\left(\frac{|f(x)|}{\lambda}\right) d\mu \leq 1 \right\}.$$

Several notable examples of Orlicz functions, along with their generated Orlicz spaces, are outlined below.

(i) Let $p \in [1, \infty]$, and define $\phi(y) = y^p$ when $p \in [1, \infty)$. For $p = \infty$, we set

$$\phi(y) = \begin{cases} 0, & \text{when } y \leq 1, \\ \infty, & \text{when } y > 1. \end{cases}$$

This generates the classical Lebesgue spaces $L^p(\mathbb{R})$, $1 \leq p \leq \infty$.

(ii) $\phi(y) = (y - 1)\chi_{[1,\infty)}(y)$, associated with the combined space $L^1(\mathbb{R}^n) + L^\infty(\mathbb{R}^n)$.

(iii) $\phi(y) = y \log y$, which defines the logarithmic space $L \log L(\mathbb{R}^n)$.

(iv) $\phi(y) = \exp(y^a) - 1$ for $a > 0$, leading to Exponential spaces $\exp L^a$.

Definition 2.4 *(Δ_2−condition)* The Δ_2−condition for an Orlicz function ϕ is satisfied if there exists a constant $K > 0$, such that

$$\phi(2u) \leq K\phi(u), \quad \forall u > 0.$$

When a function ϕ satisfies the Δ_2−condition, we will write $\phi \in \Delta_2$. The Orlicz space associated to Orlicz function with Δ_2-condition provides important regularity properties. In particular,

- For every $f \in L^\phi(\Omega)$, $I^\phi(\lambda f) < \infty$ for every $\lambda > 0$.
- Given a Banach space structure, modular convergence and norm convergence coincide.
- For $\phi, \psi \in \Delta_2$, where ψ is Orlicz conjugate of ϕ, the space $L^\phi(\Omega)$ is reflexive.

Examples of Orlicz functions that satisfy the Δ_2−condition are $\phi(u) = u(\log(e+u))$ and $\phi(u) = u^p$, for $1 \leq p < \infty$. However, $\phi(u) = (e^u - u - 1)$ does not belong to Δ_2.

Definition 2.5 *(Jensen's integral inequality)* The Jensen's integral inequality is satisfied for every function $f \in L^\phi(\Omega)$ for which the following inequality holds

$$\phi\left(\frac{1}{\mu(\Omega)}\int_\Omega f(x)\,d\mu\right) \leq \frac{1}{\mu(\Omega)}\int_\Omega \phi(f(x))\,d\mu.$$

For detailed discussions on the definitions and results concerning Orlicz spaces, see [4, 30, 37, 39, 40, 43, 45, 47, 48].

2.4 The Mixed-norm Orlicz spaces

In Subsection 2.3, we introduced classical Orlicz spaces based on an Orlicz function. We now extend this framework to the setting of mixed-norms.

Let $\overrightarrow{\Phi} = (\phi_1, \phi_2, \ldots, \phi_n)$ be a finite sequence of Orlicz functions. The modular functional $I^{\overrightarrow{\Phi}} : M(\Omega) \to [0, \infty]$ associated to $\overrightarrow{\Phi}$ is defined by

$$I^{\overrightarrow{\Phi}}(f) := \int_{\Omega_n} \phi_n \cdots \left(\int_{\Omega_2} \phi_2 \left(\int_{\Omega_1} \phi_1 \left(|f(x_1, \ldots, x_n)| \right) d\mu_1 \right) d\mu_2 \cdots \right) d\mu_n.$$

Definition 2.6 *(Mixed-norm Orlicz space)* The mixed-norm Orlicz space generated by $\overrightarrow{\Phi} = (\phi_1, \phi_2, \ldots, \phi_n)$ is defined as

$$L^{\overrightarrow{\Phi}}(\Omega) := \{ f \in M(\Omega) : I^{\overrightarrow{\Phi}}(\lambda f) < \infty \ \text{ for some } \lambda > 0 \}$$

induced with the norm

$$\|f\|_{\overrightarrow{\Phi}} = \inf \left\{ \lambda > 0 : I^{\overrightarrow{\Phi}}(f/\lambda) \le 1 \right\}.$$

The space $L^{\overrightarrow{\Phi}}(\Omega)$ generalizes the notion of mixed-norm Lebesgue space $L^{\overrightarrow{P}}(\Omega)$, as introduced in Subsection 2.2, particularly under the assumption that the components of $\overrightarrow{P} = (p_1, p_2, \ldots, p_n)$ are non-decreasing. Specifically, the Orlicz functions are chosen as $\phi_1(x) = x^{p_1}$ and $\phi_i(x) = x^{\frac{p_i}{p_{i-1}}}$, for $2 \le i \le n$, the corresponding mixed-norm Orlciz space $L^{\overrightarrow{P}}(\Omega)$ coincides with mixed-norm Lebesgue space $L^{\overrightarrow{P}}(\Omega)$.

We say that $\overrightarrow{\Phi} = (\phi_1, \ldots, \phi_n) \in \Delta_2$, if each ϕ_i satisfies Δ_2-condition. The notion of modular convergence in $L^{\overrightarrow{\Phi}}(\Omega)$ plays an important role to study the norm convergence. A sequence of functions $(f_k)_{k>0}$ in $L^{\overrightarrow{\Phi}}(\Omega)$ is said modularly convergent to a function $f \in L^{\overrightarrow{\Phi}}(\Omega)$ if

$$\lim_{k \to \infty} I^{\overrightarrow{\Phi}}(\lambda(f_k - f)) = 0$$

for some $\lambda > 0$. If $\overrightarrow{\Phi} \in \Delta_2$ then for every $\lambda > 0$, $I^{\overrightarrow{\Phi}}(\lambda(f_k - f)) \to 0$ as $k \to \infty$. Hence, $\|f_k - f\|_{\overrightarrow{\Phi}} \to 0$ as $k \to \infty$. Moreover, the norm convergence and modular convergence are equivalent if and only if $\overrightarrow{\Phi}$ satisfies the Δ_2-condition.

The Shannon sampling series (1.1) is limited by the assumption of band-limited signals [17]. To extend this study to a more general setting, Butzer et al. [11] introduced the generalized sampling series. In the subsequent section, we study the generalized sampling series within the framework of mixed-norm Lebesgue space.

3 Generalized sampling operator on mixed-norm Lebesgue space

A function is said to be a kernel $\chi : \mathbb{R}^n \to \mathbb{R}$ if it belongs to $L^1(\mathbb{R}^n)$, is bounded in a neighborhood of the origin, and fulfills the following conditions:

1. For any $\mathbf{u} \in \mathbb{R}^n$, it follows that

$$\sum_{\mathbf{k}\in\mathbb{Z}^n} \chi(\mathbf{u}-\mathbf{k}) = 1. \tag{3.1}$$

2. For some $\alpha > 0$, the absolute moment of order α is finite, i.e.,

$$m_\alpha(\chi) := \sup_{\mathbf{u}\in\mathbb{R}^n} \sum_{\mathbf{k}\in\mathbb{Z}^n} |\chi(\mathbf{u}-\mathbf{k})| \, \|\mathbf{u}-\mathbf{k}\|_2^\alpha < +\infty, \tag{3.2}$$

where $\|\cdot\|_2$ denotes the euclidean norm.

The reader can easily verify that the kernel $\chi \in L^1(\mathbb{R}^n)$ with (3.1) and (3.2) implies

$$m_0(\chi) \geq \max\{1, \|\chi\|_1\}.$$

Definition 3.1 *(Relatively separated set)* A countable set $\Gamma \subset \mathbb{R}^n$ is defined to be *relatively separated set* with gap $\kappa > 0$ if it satisfies the following conditions:

$$A_\Gamma(\kappa) := \inf_{\mathbf{x}\in\mathbb{R}^n} \sum_{\gamma\in\Gamma} \mathbf{1}_{B(\gamma;\frac{\kappa}{2})}(\mathbf{x}) \geq 1,$$
$$B_\Gamma(\kappa) := \sup_{\mathbf{x}\in\mathbb{R}^n} \sum_{\gamma\in\Gamma} \mathbf{1}_{B(\gamma;\frac{\kappa}{2})}(\mathbf{x}) < \infty,$$

where $B(\gamma; r)$ represents an open cube in a normed linear space $(\mathbb{R}^n, \|\cdot\|_\infty)$ with side-length $2r$. The notion of a relatively separated set in $\mathbb{R}$ is referred to as an admissible sequence [5].

Definition 3.2 The multivariate generalized sampling series associated with a signal f and kernel χ is defined as

$$S_w(f)(\mathbf{x}) = \sum_{\mathbf{k}\in\mathbb{Z}^n} f\left(\frac{\mathbf{k}}{w}\right) \chi(w\mathbf{x}-\mathbf{k}), \quad \mathbf{x} \in \mathbb{R}^n, \quad w > 0. \tag{3.3}$$

The multivariate generalized sampling series (3.3) does not converge in general in $L^{\vec{P}}(\mathbb{R}^n)$ [7]. Therefore, we need to restrict our discussion to an appropriate subspace of $L^{\vec{P}}(\mathbb{R}^n)$. Let us define the space $\Delta^{\vec{P}}(\mathbb{R}^n)$ by

$$\Delta^{\vec{P}}(\mathbb{R}^n) := \left\{ f \in M(\mathbb{R}^n) : \|(f(x_\mathbf{k}))\|_{\ell_w^{\vec{P}}} < \infty \text{ for each relatively separated set } (x_\mathbf{k})_{\mathbf{k}\in\mathbb{Z}^n} \right\}.$$

In the following proposition, we show that $\Delta^{\vec{P}}(\mathbb{R}^n)$ is a proper subspace of $L^{\vec{P}}(\mathbb{R}^n)$.

Proposition 3.3 *The space $\Delta^{\vec{P}}(\mathbb{R}^n)$ is a proper subspace of $L^{\vec{P}}(\mathbb{R}^n)$.*

Proof We begin by demonstrate the result for $n = 2$ and then generalize it for any $n \in \mathbb{N}$. We only need to show that $f \in \Delta^{\overrightarrow{P}}$ implies $\left(\int_{\mathbb{R}}\left(\int_{\mathbb{R}} |f(x_1, x_2)|^{p_1} dx_1\right)^{\frac{p_2}{p_1}} dx_2\right)^{\frac{1}{p_2}}$ is finite. Since f is bounded, it is possible to find $x^*_{k_1} \in [k_1, k_1+1]$, $y^*_{k_2} \in [k_2, k_2+1]$, for each $k_1, k_2 \in \mathbb{Z}$, such that

$$\operatorname*{ess\,sup}_{x_1\in[k_1,k_1+1],\, x_2\in[k_2,k_2+1]} |f(x_1, x_2)| < |f(x^*_{k_1}, y^*_{k_2})| + \frac{1}{1 + k_1^2 + k_2^2},$$

and it follows that

$$\begin{aligned}
\|f\|_{(p_1,p_2)} &= \left(\sum_{k_2\in\mathbb{Z}} \int_{k_2}^{k_2+1} \left(\sum_{k_1\in\mathbb{Z}} \int_{k_1}^{k_1+1} |f(x_1, x_2)|^{p_1} dx_1\right)^{\frac{p_2}{p_1}} dx_2\right)^{\frac{1}{p_2}}\\
&< \left(\sum_{k_2\in\mathbb{Z}} \left(\sum_{k_1\in\mathbb{Z}} \left(|f(x^*_{k_1}, y^*_{k_2})| + \frac{1}{1 + k_1^2 + k_2^2}\right)^{p_1}\right)^{\frac{p_2}{p_1}}\right)^{\frac{1}{p_2}}\\
&\leq \left(\sum_{k_2\in\mathbb{Z}} \left(\sum_{k_1\in\mathbb{Z}} |f(x^*_{k_1}, y^*_{k_2})|^{p_1}\right)^{\frac{p_2}{p_1}}\right)^{\frac{1}{p_2}} + \left(\sum_{k_2\in\mathbb{Z}} \left(\sum_{k_1\in\mathbb{Z}} \left(\frac{1}{1 + k_1^2 + k_2^2}\right)^{p_1}\right)^{\frac{p_2}{p_1}}\right)^{\frac{1}{p_2}}\\
&:= S_1 + S_2\,.
\end{aligned}$$

Indeed, S_2 is finite for each $p_1, p_2 \geq 1$. Concerning S_1, the problem is that $(x^*_{k_1}, y^*_{k_2})_{(k_1,k_2)\in\mathbb{Z}^2}$ may not necessarily be a relatively separated in $\mathbb{R}^2$, but the subsequences $(x^*_{2k_1}, y^*_{2k_2})_{(k_1,k_2)\in\mathbb{Z}^2}$, $(x^*_{2k_1+1}, y^*_{2k_2})_{(k_1,k_2)\in\mathbb{Z}^2}$, $(x^*_{2k_1}, y^*_{2k_2+1})_{(k_1,k_2)\in\mathbb{Z}^2}$, $(x^*_{2k_1+1}, y^*_{2k_2+1})_{(k_1,k_2)\in\mathbb{Z}^2}$ are relatively separated set in $\mathbb{R}^2$, and hence for each $f \in \Delta^{\overrightarrow{P}}$, we have

$$\begin{aligned}
&\left(\sum_{k_2\in\mathbb{Z}} \left(\sum_{k_1\in\mathbb{Z}} |f(x^*_{k_1}, y^*_{k_2})|^{p_1}\right)^{\frac{p_2}{p_1}}\right)^{\frac{1}{p_2}}\\
\leq&\left(\sum_{k_2\in\mathbb{Z}} \left(\sum_{k_1\in\mathbb{Z}} |f(x^*_{2k_1}, y^*_{2k_2})|^{p_1}\right)^{\frac{p_2}{p_1}}\right)^{\frac{1}{p_2}} + \left(\sum_{k_2\in\mathbb{Z}} \left(\sum_{k_1\in\mathbb{Z}} |f(x^*_{2k_1}, y^*_{2k_2+1})|^{p_1}\right)^{\frac{p_2}{p_1}}\right)^{\frac{1}{p_2}}\\
&+ \left(\sum_{k_2\in\mathbb{Z}} \left(\sum_{k_1\in\mathbb{Z}} |f(x^*_{2k_1+1}, y^*_{2k_2})|^{p_1}\right)^{\frac{p_2}{p_1}}\right)^{\frac{1}{p_2}} + \left(\sum_{k_2\in\mathbb{Z}} \left(\sum_{k_1\in\mathbb{Z}} |f(x^*_{2k_1+1}, y^*_{2k_2+1})|^{p_1}\right)^{\frac{p_2}{p_1}}\right)^{\frac{1}{p_2}}\\
<&\infty.
\end{aligned}$$

In a similar manner, for $n \in \mathbb{N}$ there exist 2^n subsequent relatively separated sets. Using the same reasoning, since $f \in \Delta^{\overrightarrow{P}}(\mathbb{R}^n)$, it follows that the all the mixed-norm Lebesgue space are finite. Hence, $f \in L^{\overrightarrow{P}}(\mathbb{R}^n)$. Consider

$$f(\mathbf{x}) = \begin{cases} m, & \text{if } \mathbf{x} = \mathbf{x}_m,\\ 0, & \text{if } otherwise, \end{cases}$$

where $\{\mathbf{x}_m : m \in \mathbb{N}\} \subset \mathbb{R}^n$ is a countable set. Since the support of f is countable, we have $f \in L^{\overrightarrow{P}}(\mathbb{R}^n)$, while for a suitable relatively separated set $(x_{\mathbf{k}})_{\mathbf{k}\in\mathbb{Z}^n}$ containing all support points, the discrete mixed-norm $\|(f(x_{\mathbf{k}}))\|_{\ell_w^{\overrightarrow{P}}}$ is not finite, so $f \notin \Delta^{\overrightarrow{P}}(\mathbb{R}^n)$.

This completes the proof. □

By restricting the allowable functions f in (3.3) to the space $\Delta^{\overrightarrow{P}}$, the operators $(S_w)_{w>0}$ becomes a bounded operator from $\Delta^{\overrightarrow{P}}$ into $L^{\overrightarrow{P}}$.

Theorem 3.4 *The generalized sampling operator* $S_w : \Delta^{\overrightarrow{P}}\left(\subset L^{\overrightarrow{P}}(\mathbb{R}^n)\right) \to L^{\overrightarrow{P}}(\mathbb{R}^n)$ *satisfies*

$$\|S_w f\|_{\overrightarrow{P}} \leq (m_0(\chi))^{1-\frac{1}{p_n}} \|\chi\|_1^{\frac{1}{p_n}} \left\| f\left(\frac{\boldsymbol{k}}{w}\right)\right\|_{\ell_w^{\overrightarrow{P}}}.$$

Proof To begin, we verifying the result for $n = 2$. Consider $\overrightarrow{P} = (p_1, p_2) \in [1,\infty)^2$ and $f \in L^{\overrightarrow{P}}(\mathbb{R}^2)$. From convexity property on the summation and Jensen's inequality on integration, we get

$$\begin{aligned}
&\int_{\mathbb{R}} \left| \sum_{k_2\in\mathbb{Z}} \sum_{k_1\in\mathbb{Z}} |\chi(wx_1 - k_1, wx_2 - k_2)| f\left(\frac{k_1}{w}, \frac{k_2}{w}\right)\right|^{p_1} dx_1 \\
&\leq (m_0(\chi))^{p_1} \int_{\mathbb{R}} \sum_{k_2\in\mathbb{Z}} \sum_{k_1\in\mathbb{Z}} \frac{|\chi(wx_1 - k_1, wx_2 - k_2)|}{m_0(\chi)} \left| f\left(\frac{k_1}{w}, \frac{k_2}{w}\right)\right|^{p_1} dx_1 \\
&\leq \frac{(m_0(\chi))^{p_1-1}}{w} \sum_{k_2\in\mathbb{Z}} \sum_{k_1\in\mathbb{Z}} \|\chi(\cdot, wx_2 - k_2)\|_1 \left| f\left(\frac{k_1}{w}, \frac{k_2}{w}\right)\right|^{p_1}.
\end{aligned}$$

Since $p_1 \leq p_2$, therefore using the convexity and Jensen's inequality on integration, we get

$$\begin{aligned}
&\|S_w f\|_{(p_1,p_2)}^{p_2} \\
&\leq m_0(\chi)^{p_2-\frac{p_2}{p_1}} \int_{\mathbb{R}} \left(\sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2 - k_2)\|_1 \right)^{\frac{p_2}{p_1}} \\
&\quad \times \left(\frac{1}{w} \sum_{k_2\in\mathbb{Z}} \frac{\|\chi(\cdot, wx_2 - k_2)\|_1}{\sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2 - k_2)\|_1} \sum_{k_1\in\mathbb{Z}} \left| f\left(\frac{k_1}{w}, \frac{k_2}{w}\right)\right|^{p_1} \right)^{\frac{p_2}{p_1}} dx_2 \\
&\leq m_0(\chi)^{p_2-\frac{p_2}{p_1}} \int_{\mathbb{R}} \left(\sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2 - k_2)\|_1 \right)^{\frac{p_2}{p_1}-1} \\
&\quad \times \sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2 - k_2)\|_1 \left(\frac{1}{w} \sum_{k_1\in\mathbb{Z}} \left| f\left(\frac{k_1}{w}, \frac{k_2}{w}\right)\right|^{p_1} \right)^{\frac{p_2}{p_1}} dx_2.
\end{aligned}$$

Since the absolute moment of the kernel χ of order zero is finite, for every $1 \leq j \leq n$, we have

$$\begin{aligned}
&\sum_{k_n\in\mathbb{Z}} \cdots \sum_{k_j\in\mathbb{Z}} \|\chi(\cdot, \cdots, \cdot, wx_j - k_j, \ldots, wx_n - k_n)\|_1 \\
&\quad \leq \int_{[0,1]^{j-1}} \sum_{\mathbf{k}\in\mathbb{Z}^n} |\chi(x_1 - k_1, x_2 - k_2, \ldots, wx_n - k_n)| dx_1 dx_2 \cdots dx_{j-1} \\
&\quad \leq \sup_{\mathbf{x}\in\mathbb{R}^n} \sum_{\mathbf{k}\in\mathbb{Z}^n} |\chi(\mathbf{x} - \mathbf{k})| = m_0(\chi).
\end{aligned}$$

Hence,

$$\begin{aligned}
\|S_w f\|_{(p_1,p_2)}^{p_2} &\leq \frac{m_0(\chi)^{p_2-1}}{w} \|\chi\|_1 \sum_{k_2\in\mathbb{Z}} \left(\frac{1}{w} \sum_{k_1\in\mathbb{Z}} \left| f\left(\frac{k_1}{w}, \frac{k_2}{w}\right) \right|^{p_1} \right)^{\frac{p_2}{p_1}} \\
&\leq m_0(\chi)^{p_2-1} \|\chi\|_1 \left\| f\left(\frac{k_1}{w}, \frac{k_2}{w}\right) \right\|_{\ell_w^{\overrightarrow{P}}}^{p_2} \\
\|S_w f\|_{(p_1,p_2)} &\leq (m_0(\chi))^{1-\frac{1}{p_2}} \|\chi\|_1^{\frac{1}{p_2}} \left\| f\left(\frac{k_1}{w}, \frac{k_2}{w}\right) \right\|_{\ell_w^{\overrightarrow{P}}}.
\end{aligned}$$

Now following the similar lines of proof for $\overrightarrow{P} = (p_1, p_2, \ldots, p_n)$, we get

$$\|S_w f\|_{\overrightarrow{P}} \leq (m_0(\chi))^{1-\frac{1}{p_n}} \|\chi\|_1^{\frac{1}{p_n}} \left\| f\left(\frac{\mathbf{k}}{w}\right) \right\|_{\ell_w^{\overrightarrow{P}}}.$$

This completes the proof. □

In the next section, we will discuss the well-known family of Kantorovich-type sampling operators. These operators play a significant role in the approximation of not-necessarily continuous functions, with particular importance in multivariate contexts. Moreover, this family of operators has been proven to be especially effective in the field of image processing.

4 Kantorovich-type sampling operators on mixed-norm Lebesgue space

In this section, we investigate the boundedness and convergence of Kantorovich-type sampling operators within mixed-norm Lebesgue space. It is noteworthy that the convergence of $(K_w)_{w>0}$ follows from the fact that $C_c(\mathbb{R}^n)$ is dense in $L^{\overrightarrow{P}}(\mathbb{R}^n)$. First we proceed to demonstrate that for a fixed $w > 0$, the operator $K_w : L^{\overrightarrow{P}}(\mathbb{R}^n) \to L^{\overrightarrow{P}}(\mathbb{R}^n)$ is a bounded linear operator.

Definition 4.1 Let $f:\mathbb{R}^n\to\mathbb{R}$ be a locally integrable function. For $\mathbf{x}\in\mathbb{R}^n$, the Kantorovich-type sampling series associated with f is given by (see [25])

$$K_w(f)(\mathbf{x})=\sum_{\mathbf{k}\in\mathbb{Z}^n}\chi(w\mathbf{x}-\mathbf{k})\ w^n\int_{I_{\mathbf{k},w}}f(\mathbf{t})d\mathbf{t} \tag{4.1}$$

where $I_{\mathbf{k},w}=\prod_{j=1}^{n}\left[\frac{k_j}{w},\frac{k_j+1}{w}\right]$.

Theorem 4.2 *The Kantorovich-type sampling operator* $K_w:L^{\overrightarrow{P}}(\mathbb{R}^n)\to L^{\overrightarrow{P}}(\mathbb{R}^n)$ *satisfies*

$$\|K_wf\|_{\overrightarrow{P}}\le(m_0(\chi))^{1-\frac{1}{p_n}}\ \|\chi\|_1^{\frac{1}{p_n}}\ \|f\|_{\overrightarrow{P}}\,.$$

Proof We first examine this result for $n=2$, which will further be extend to any $n\in\mathbb{N}$. Now using (4.1), we obtain

$$\begin{aligned}&\|K_wf\|_{(p_1,p_2)}^{p_2}\\&=\int_{\mathbb{R}}\left(\int_{\mathbb{R}}\left|\sum_{k_2\in\mathbb{Z}}\sum_{k_1\in\mathbb{Z}}|\chi(wx_1-k_1,wx_2-k_2)|w^2\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}}\int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}}f(t_1,t_2)dt_1dt_2\right|^{p_1}dx_1\right)^{\frac{p_2}{p_1}}dx_2.\end{aligned}$$

From convexity property on the summation and Jensen's inequality on integration, we get

$$\begin{aligned}&\int_{\mathbb{R}}\left|\sum_{k_2\in\mathbb{Z}}\sum_{k_1\in\mathbb{Z}}|\chi(wx_1-k_1,wx_2-k_2)|w^2\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}}\int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}}|f(t_1,t_2)|dt_1dt_2\right|^{p_1}dx_1\\&\le(m_0(\chi))^{p_1}\int_{\mathbb{R}}\sum_{k_2\in\mathbb{Z}}\sum_{k_1\in\mathbb{Z}}\frac{|\chi(wx_1-k_1,wx_2-k_2)|}{m_0(\chi)}\left(w^2\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}}\int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}}|f(t_1,t_2)|dt_1dt_2\right)^{p_1}\\&\le(m_0(\chi))^{p_1-1}\sum_{k_2\in\mathbb{Z}}\sum_{k_1\in\mathbb{Z}}\|\chi(\cdot,wx_2-k_2)\|_1\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}}\int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}}w|f(t_1,t_2)|^{p_1}dt_1dt_2\\&\le(m_0(\chi))^{p_1-1}\sum_{k_2\in\mathbb{Z}}\|\chi(\cdot,wx_2-k_2)\|_1\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}}\int_{\mathbb{R}}w|f(t_1,t_2)|^{p_1}dt_1dt_2.\end{aligned}$$

Since $p_1\le p_2$, using the convexity and Jensen's inequality on integration, we get

$$\begin{aligned}&\|K_wf\|_{(p_1,p_2)}^{p_2}\\&\le m_0(\chi)^{p_2-\frac{p_2}{p_1}}\int_{\mathbb{R}}\left(\sum_{k_2\in\mathbb{Z}}\|\chi(\cdot,wx_2-k_2)\|_1\right)^{\frac{p_2}{p_1}}\\&\quad\times\left(\sum_{k_2\in\mathbb{Z}}\frac{\|\chi(\cdot,wx_2-k_2)\|_1}{\sum_{k_2\in\mathbb{Z}}\|\chi(\cdot,wx_2-k_2)\|_1}w\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}}\int_{\mathbb{R}}|f(t_1,t_2)|^{p_1}dt_1dt_2\right)^{\frac{p_2}{p_1}}dx_2\end{aligned}$$

$$\leq m_0(\chi)^{p_2-\frac{p_2}{p_1}} \int_{\mathbb{R}} \left(\sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2-k_2)\|_1 \right)^{\frac{p_2}{p_1}-1}$$

$$\times \sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2-k_2)\|_1 \left(w \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\mathbb{R}} |f(t_1,t_2)|^{p_1} dt_1 dt_2 \right)^{\frac{p_2}{p_1}} dx_2$$

$$\leq m_0(\chi)^{p_2-1} \|\chi\|_1 \sum_{k_2\in\mathbb{Z}} \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \left(\int_{\mathbb{R}} |f(t_1,t_2)|^{p_1} dt_1 \right)^{\frac{p_2}{p_1}} dt_2$$

$$\leq m_0(\chi)^{p_2-1} \|\chi\|_1 \|f\|_{(p_1,p_2)}^{p_2}.$$

This gives

$$\|K_w f\|_{(p_1,p_2)} \leq (m_0(\chi))^{1-\frac{1}{p_2}} \|\chi\|_1^{\frac{1}{p_2}} \|f\|_{(p_1,p_2)}.$$

Proceeding in a similar manner, we obtain

$$\|K_w f\|_{\overrightarrow{P}} \leq (m_0(\chi))^{1-\frac{1}{p_n}} \|\chi\|_1^{\frac{1}{p_n}} \|f\|_{\overrightarrow{P}}.$$

□

For a given $f \in L^{\overrightarrow{P}}(\mathbb{R}^n)$, the Kantorovich-type sampling operator K_w acting on f approximates the function f in $L^{\overrightarrow{P}}(\mathbb{R}^n)$. In order to prove this result, we first observe that $C_c(\mathbb{R}^n)$ is dense in $L^{\overrightarrow{P}}(\mathbb{R}^n)$ and $K_w f \to f$ as $w \to \infty$ for every $f \in C_c(\mathbb{R}^n) \subset L^{\overrightarrow{P}}(\mathbb{R}^n)$. In order to prove the convergence result in $L^{\overrightarrow{P}}(\mathbb{R}^n)$, the following lemma on the absolute continuity of χ plays a key role.

Lemma 4.3 *Let χ be the kernel. For every $c > 0$ and $\epsilon > 0$, there exists a constant $L > 0$ such that*

$$\int_{\|\boldsymbol{x}\|_\infty > L} w^n |\chi(w\boldsymbol{x}-\boldsymbol{k})| d\boldsymbol{x} < \epsilon$$

for sufficiently large $w > 0$ and $\boldsymbol{k} \in \mathbb{Z}^n$ such that $\|\boldsymbol{k}\|_\infty \leq wc$.

Proof Since $\chi \in L^1(\mathbb{R}^n)$, there exists a constant $M > 0$ such that

$$\int_{\|\mathbf{u}\|_\infty > M} |\chi(\mathbf{u})| d\mathbf{u} < \epsilon.$$

For $c > 0$, $w \geq 1$, consider $\mathbf{k}$ such that $\|\mathbf{k}\|_\infty \leq wc$. Further we can also find $L > 0$ such that $w(L-c) > M$. The detailed proof can be found in [25]. □

In the following theorem, we demonstrate the convergence of the Kantorovich-type sampling operator in $C_c(\mathbb{R}^n)$ within the framework of mixed-norm Lebesgue space.

Theorem 4.4 *For $f \in C_c(\mathbb{R}^n)$, it holds that*

$$\lim_{w\to\infty} \|K_w f - f\|_{\overrightarrow{P}} = 0.$$

Proof We proceed to analyze the result for $n=2$, i.e., for $f \in C_c(\mathbb{R}^2)$, and afterwards we will generalize it for any $n \in \mathbb{N}$. We will prove that

$$\lim_{w\to\infty} \|K_w f - f\|_{(p_1,p_2)} = 0.$$

We will be using the *Vitali Convergence theorem* in mixed-norm Lebesgue space [31]. We outline the proof in the following steps:

Step 1. In view of [25, Theorem 4.1], we have

$$\lim_{w\to\infty} \|K_w f - f\|_\infty = 0.$$

Step 2. As $f \in C_c(\mathbb{R}^n)$, there exists $a>0$ such that $\operatorname{supp} f \subseteq [-a,a]^2$. For every sufficiently large $w>0$, whenever $(k_1,k_2) \notin [-wa-1, wa]^2$, we have $\left\{[\frac{k_1}{w}, \frac{k_1+1}{w}] \times [\frac{k_2}{w}, \frac{k_2+1}{w}]\right\} \cap [-a,a]^2 = \emptyset$. This gives

$$\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)| dt_1 dt_2 = 0.$$

By Lemma 4.3, for every fixed $\epsilon>0$, there exists a constant $M>0$ (we can assume $M>a$) such that

$$\int_{\|\mathbf{x}\|_\infty > M} w^n |\chi(w\mathbf{x}-\mathbf{k})| d\mathbf{x} < \epsilon$$

for every $(k_1,k_2) \in [-wa-1, wa]^2$. For a set $\mathbf{G} = [-M,M]^2$, we compute

$$\begin{aligned}
I &= \|K_w f\|^{p_2}_{L^{(p_1,p_2)}(\mathbf{G}^c)} \\
&\leq \left(\int_{|x_2|>M} \left(\int_{|x_1|\leq M} |(K_w f)(x_1,x_2)|^{p_1} dx_1 \right)^{\frac{p_2}{p_1}} dx_2 \right) \\
&\quad + \left(\int_{|x_2|\leq M} \left(\int_{|x_1|> M} |(K_w f)(x_1,x_2)|^{p_1} dx_1 \right)^{\frac{p_2}{p_1}} dx_2 \right) \\
&\quad + \left(\int_{|x_2|> M} \left(\int_{|x_1|> M} |(K_w f)(x_1,x_2)|^{p_1} dx_1 \right)^{\frac{p_2}{p_1}} dx_2 \right) \\
&:= I_1 + I_2 + I_3.
\end{aligned} \tag{4.2}$$

First we compute I_1. Utilizing Jensen's inequality twice and Fubini-Tonelli theorem, we have

$$\begin{aligned}
&\int_{|x_1|\leq M} \left(\sum_{\frac{k_2}{w} \in [-a-\frac{1}{w}, a]} \sum_{\frac{k_1}{w} \in [-a-\frac{1}{w}, a]} |\chi(wx_1-k_1, wx_2-k_2)| w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)| dt_1 dt_2 \right)^{p_1} dx_1 \\
&\leq (m_0(\chi))^{p_1-1} \sum_{\frac{k_2}{w} \in [-a-\frac{1}{w}, a]} \sum_{\frac{k_1}{w} \in [-a-\frac{1}{w}, a]} w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)|^{p_1} dt_1 dt_2
\end{aligned}$$

$$
\begin{aligned}
&\times \int_{|x_1|\le M} |\chi(wx_1-k_1, wx_2-k_2)|dx_1\\
&\le \left(m_0(\chi)\right)^{p_1-1} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M])}\\
&\times w\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} |f(t_1,t_2)|^{p_1} dt_1 dt_2.
\end{aligned}
$$

Again using Jensen's inequality and Fubini-Tonelli theorem, we get

$$
\begin{aligned}
I_1 &\le \left(m_0(\chi)\right)^{p_2-\frac{p_2}{p_1}} \int_{|x_2|>M} \left(\sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M])} \right)^{\frac{p_2}{p_1}-1}\\
&\quad\times \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M])}\\
&\quad\times \left(w\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} |f(t_1,t_2)|^{p_1} dt_1 dt_2 \right)^{\frac{p_2}{p_1}} dx_2\\
&\le \left(m_0(\chi)\right)^{p_2-1} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \left(\int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} |f(t_1,t_2)|^{p_1} dt_1 \right)^{\frac{p_2}{p_1}} dt_2\\
&\quad\times \int_{|x_2|>M} w\|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M])} dx_2.
\end{aligned}
$$

Thus

$$
I_1 < \frac{\epsilon}{3}\,\left(m_0(\chi)\right)^{p_2-1} \|f\|_{(p_1,p_2)}^{p_2}.
$$

Note that the last inequality follows from the following observation:

$$
\int_{|x_2|>M} w\|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M])}\, dx_2 < \frac{\epsilon}{3}.
$$

Similarly, we can compute I_2. Thus we get

$$
I_2 < \frac{\epsilon}{3}\,\left(m_0(\chi)\right)^{p_2-1} \|f\|_{(p_1,p_2)}^{p_2}.
$$

One can wite I_3 as follows:

$$
I_3 := \Bigg(\int_{|x_2|>M} \Bigg(\int_{|x_1|>M} \Bigg(\sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \sum_{\frac{k_1}{w}\in[-a-\frac{1}{w},a]} |\chi(wx_1-k_1, wx_2-k_2)|
$$

$$w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)| dt_1 dt_2\Bigg)^{p_1} dx_1\Bigg)^{\frac{p_2}{p_1}} dx_2\Bigg).$$

In view of Jensen's inequality and Fubini-Tonelli theorem, we have

$$\begin{aligned}
&\int_{|x_1|>M} \Bigg(\sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \sum_{\frac{k_1}{w}\in[-a-\frac{1}{w},a]} |\chi(wx_1-k_1, wx_2-k_2)| w^2 \\
&\quad\times \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)| dt_1 dt_2\Bigg)^{p_1} dx_1 \\
&\quad\leq \big(m_0(\chi)\big)^{p_1-1} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \sum_{\frac{k_1}{w}\in[-a-\frac{1}{w},a]} \Bigg(w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)| dt_1 dt_2\Bigg)^{p_1} \\
&\quad\times \int_{|x_1|>M} |\chi(wx_1-k_1, wx_2-k_2)| dx_1 \\
&\leq \big(m_0(\chi)\big)^{p_1-1} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \sum_{\frac{k_1}{w}\in[-a-\frac{1}{w},a]} w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)|^{p_1} dt_1 dt_2 \\
&\quad\times \int_{|x_1|>M} |\chi(wx_1-k_1, wx_2-k_2)| dx_1 \\
&\quad\leq \big(m_0(\chi)\big)^{p_1-1} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M]^c)} \\
&\quad w \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} |f(t_1,t_2)|^{p_1} dt_1 dt_2.
\end{aligned}$$

Again using Jensen's inequality and Fubini-Tonelli theorem, we obtain

$$\begin{aligned}
I_3 &\leq \big(m_0(\chi)\big)^{p_2-\frac{p_2}{p_1}} \int_{|x_2|>M} \Bigg(\sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M]^c)} \Bigg)^{\frac{p_2}{p_1}-1} \\
&\quad\times \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M]^c)} \\
&\quad\times \Bigg(w \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} |f(t_1,t_2)|^{p_1} dt_1 dt_2 \Bigg)^{\frac{p_2}{p_1}} dx_2 \\
&\leq \big(m_0(\chi)\big)^{p_2-1} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \Bigg(\int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} |f(t_1,t_2)|^{p_1} dt_1 \Bigg)^{\frac{p_2}{p_1}} dt_2
\end{aligned}$$

$$\times \int_{|x_2|>M} w \| \chi(\cdot, wx_2 - k_2) \|_{L^1([-M,M]^c)} dx_2.$$

In view of Lemma 4.3, we obtain

$$I_3 < \frac{\epsilon}{3} \left(m_0(\chi)\right)^{p_2-1} \|f\|_{(p_1,p_2)}^{p_2}.$$

From (4.2), we get

$$\begin{aligned} I &< \frac{\epsilon}{3} \left(m_0(\chi)\right)^{p_2-1} \|f\|_{(p_1,p_2)}^{p_2} + \frac{\epsilon}{3} \left(m_0(\chi)\right)^{p_2-1} \|f\|_{(p_1,p_2)}^{p_2} + \frac{\epsilon}{3} \left(m_0(\chi)\right)^{p_2-1} \|f\|_{(p_1,p_2)}^{p_2} \\ &< \epsilon \left(m_0(\chi)\right)^{p_2-1} \|f\|_{(p_1,p_2)}^{p_2}. \end{aligned}$$

Step 3. For every $\epsilon > 0$, $\exists\, \delta > 0$ such that

$$\left(\int_{B_2} \int_{B_1} |\chi(u_1, u_2)| du_1 du_2\right) < \frac{\epsilon^{p_2}}{(m_0(\chi))^{p_2-1} \|f\|_{(p_1,p_2)}^{p_2}} \tag{4.3}$$

for every pair of measurable sets $B_i \subset \mathbb{R}$ with $\mu(B_i) < \delta$, for $i = 1, 2$. By following the approach used in the proof of Theorem 4.2 and absolute continuity property of the Lebesgue integral (4.3), we get

$$\begin{aligned} \left(\int_{B_2} \left(\int_{B_1} |(K_w f)(x_1, x_2)|^{p_1} dx_1\right)^{\frac{p_2}{p_1}} dx_2\right)^{\frac{1}{p_2}} &\le (m_0(\chi))^{1-\frac{1}{p_2}} \|f\|_{(p_1,p_2)} \|\chi\|_{L^{(1,1)}(B_1 \times B_2)}^{\frac{1}{p_2}} \\ &< \epsilon. \end{aligned}$$

Thus, the integrals $\left(\int_{B_2} \left(\int_{B_1} |K_w f|^{p_1} dx_1\right)^{\frac{p_2}{p_1}} dx_2\right)^{\frac{1}{p_2}}$ are equi-absolutely continuous for arbitrary choices of B_1, B_2.

Hence, by using the Vitali convergence theorem, we get $\|K_w f - f\|_{(p_1,p_2)} \to 0$ as $w \to \infty$. Following along the similar lines, one can generalize the result for $n \ge 3$. □

The space of compactly supported continuous functions $C_c(\mathbb{R}^n)$ is dense in the mixed-norm Lebesgue spaces $L^{\overrightarrow{P}}(\mathbb{R}^n)$ (see [2]). Therefore, Theorem 4.4 can be generalized to a mixed-norm Lebesgue space $L^{\overrightarrow{P}}(\mathbb{R}^n)$. In particular, we have the following result.

Theorem 4.5 *For $f \in L^{\overrightarrow{P}}(\mathbb{R}^n)$, we have*

$$\lim_{w \to \infty} \|K_w f - f\|_{\overrightarrow{P}} = 0.$$

Proof Let $f \in L^{\overrightarrow{P}}(\mathbb{R}^n)$ and $\epsilon > 0$ be fixed. Using denisty of $C_c(\mathbb{R}^n)$ in $L^{\overrightarrow{P}}(\mathbb{R}^n)$, there exists a function $g \in C_c(\mathbb{R}^n)$ such that

$$\|g - f\|_{\overrightarrow{P}} < \frac{\epsilon}{2\left(1 + (m_0(\chi))^{1-\frac{1}{p_n}} \|\chi\|_1^{\frac{1}{p_n}}\right)}.$$

Utilizing the triangle inequality for $\|\cdot\|_{\overrightarrow{P}}$ and Theorem 4.2 and 4.4, it follows that

$$\begin{aligned}
\|K_w f - f\|_{\overrightarrow{P}} &\leq \|g - f\|_{\overrightarrow{P}} + \|K_w g - g\|_{\overrightarrow{P}} + \|K_w f - K_w g\|_{\overrightarrow{P}} \\
&\leq \|g - f\|_{\overrightarrow{P}} + \|K_w g - g\|_{\overrightarrow{P}} + (m_0(\chi))^{1-\frac{1}{p_n}} \|\chi\|_1^{\frac{1}{p_n}} \|g - f\|_{\overrightarrow{P}} \\
&\leq \left(1 + (m_0(\chi))^{1-\frac{1}{p_n}} \|\chi\|_1^{\frac{1}{p_n}}\right) \|g - f\|_{\overrightarrow{P}} + \|K_w g - g\|_{\overrightarrow{P}} \\
&< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\
&= \epsilon,
\end{aligned}$$

$\forall\, w \geq N_0$, for some $N_0 \in \mathbb{N}$. □

5 Kantorovich-type sampling operators on mixed-norm orlicz space

The objective of this section is to establish the convergence of Kantorovich-type sampling operators in the setting of mixed-norm Orlicz spaces $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$. To prove this result, we utilize the boundedness of the operator (4.1) and the denseness property of $C_c(\mathbb{R}^n)$ within the framework of $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$. In the following theorem, we discuss the boundedness of these operators in $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$ using the modular functional $I^{\overrightarrow{\Phi}}$.

Theorem 5.1 *Let $f \in L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$ with $\overrightarrow{\Phi} = (\phi_1, \ldots, \phi_n)$ and $\lambda > 0$, there holds*

$$I^{\overrightarrow{\Phi}}(\lambda K_w f) \leq \frac{\|\chi\|_1}{\left(m_0(\chi)\right)^n} I^{\overrightarrow{\Phi}}\left(\lambda \left(m_0(\chi)\right)^n f\right). \tag{5.1}$$

Proof First, we examine the result for $n = 2$, i.e., for $\overrightarrow{\Phi} = (\phi_1, \phi_2)$. Assume that the modular term $I^{\overrightarrow{\Phi}}-$ on the right hand side of (5.1) is finite.

From convexity property on the summation and Jensen's inequality on integration, we get

$$\int_{\mathbb{R}} \phi_1\left(\lambda \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}} |\chi(wx_1 - k_1, wx_2 - k_2)| w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1, t_2)| dt_1 dt_2\right) dx_1$$

$$\leq \frac{1}{m_0(\chi)} \int_{\mathbb{R}} \sum_{k_2\in\mathbb{Z}} \sum_{k_1\in\mathbb{Z}} \frac{|\chi(wx_1-k_1, wx_2-k_2)|}{m_0(\chi)} \phi_1$$
$$\times \left(\lambda m_0(\chi) w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)| dt_1 dt_2 \right)$$
$$\leq \frac{1}{m_0(\chi)} \sum_{k_2\in\mathbb{Z}} \sum_{k_1\in\mathbb{Z}} \|\chi(\cdot, wx_2-k_2)\|_1 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} w\phi_1\left(\lambda m_0(\chi)|f(t_1,t_2)|\right) dt_1 dt_2$$
$$\leq \frac{1}{m_0(\chi)} \sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2-k_2)\|_1 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\mathbb{R}} w\phi_1\left(\lambda m_0(\chi)|f(t_1,t_2)|\right) dt_1 dt_2.$$

Using the convexity and Jensen's inequality on integration, we get

$$I^{\vec{\Phi}}(\lambda K_w f)$$
$$\leq \frac{1}{m_0(\chi)} \int_{\mathbb{R}} \phi_2 \left(\lambda \sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2-k_2)\|_1 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\mathbb{R}} w\phi_1\left(\lambda m_0(\chi)|f(t_1,t_2)|\right) dt_1 dt_2 \right) dx_2$$
$$\leq \frac{1}{(m_0(\chi))^2} \int_{\mathbb{R}} \sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2-k_2)\|_1 \phi_2 \left(\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\mathbb{R}} w\phi_1\left(\lambda \left(m_0(\chi)\right)^2 |f(t_1,t_2)|\right) dt_1 dt_2 \right) dx_2$$
$$\leq \frac{1}{(m_0(\chi))^2} \int_{\mathbb{R}} \sum_{k_2\in\mathbb{Z}} \|\chi(\cdot, wx_2-k_2)\|_1 \, w \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \phi_2 \left(\int_{\mathbb{R}} \phi_1\left(\lambda \left(m_0(\chi)\right)^2 |f(t_1,t_2)|\right) dt_1 \right) dt_2 \, dx_2$$
$$\leq \frac{\|\chi\|_1}{(m_0(\chi))^2} I^{(\phi_1,\phi_2)}(\lambda \left(m_0(\chi)\right)^2 f).$$

Now following along the same lines, we obtain

$$I^{\vec{\Phi}}(\lambda K_w f) \leq \frac{\|\chi\|_1}{(m_0(\chi))^n} \; I^{\vec{\Phi}}(\lambda \left(m_0(\chi)\right)^n f).$$

This proves the required estimate. □

Remark 5.2 The modular inequality on Kantorovich-type sampling operator can be written in terms of norm with respect to mixed-norm Orlicz space. Let $f \in L^{\vec{\Phi}}(\mathbb{R}^n)$ be arbitrary and $\lambda = \|\left(m_0(\chi)\right)^n f\|_{\vec{\Phi}}$, then by the definition of mixed-norm Orlicz space we have

$$I^{\vec{\Phi}}\left(\frac{\left(m_0(\chi)\right)^n f}{\lambda}\right) \leq 1.$$

The inequality (5.1) implies

$$I^{\vec{\Phi}}\left(\frac{K_w f}{\lambda}\right) \leq \frac{\|\chi\|_1}{(m_0(\chi))^n} I^{\vec{\Phi}}\left(\frac{\left(m_0(\chi)\right)^n f}{\lambda}\right) \leq 1.$$

The last inequality hold from the fact that $m_0(\chi) \geq \max\{1, \|\chi\|_1\}$. Therefore, by the definition of mixed-norm Orlicz space, we have

$$\|K_w f\|_{\overrightarrow{\Phi}} \leq \| \left(m_0(\chi)\right)^n f\|_{\overrightarrow{\Phi}} \leq \left(m_0(\chi)\right)^n \|f\|_{\overrightarrow{\Phi}}.$$

The following theorem establishes the convergence of $(K_w)_{w>0}$ in $C_c(\mathbb{R}^n)$ considered within the framework of mixed-norm Orlicz spaces.

Theorem 5.3 *For* $\overrightarrow{\Phi} = (\phi_1, \ldots, \phi_n)$, $\lambda > 0$ *and* $f \in C_c(\mathbb{R}^n)$*, we have*

$$\lim_{w\to\infty} I^{\overrightarrow{\Phi}}(\lambda(K_w f - f)) = 0.$$

Proof We begin with $\overrightarrow{\Phi} = (\phi_1, \phi_2)$ for better mathematical visualization. Here we prove that

$$\lim_{w\to\infty} I^{\overrightarrow{\Phi}}(\lambda(K_w f - f)) = 0,$$

for every $\lambda > 0$. We will be using the *Vitali Convergence theorem* for mixed-norm space [31]. The proof proceeds in the following steps:

Step 1. Utilizing [25, Theorem 4.1], we get

$$\lim_{w\to\infty} \phi_2(\phi_1(\lambda\|K_w f - f\|_\infty)) = 0.$$

Step 2. As $f \in C_c(\mathbb{R}^n)$, i.e., supp $f \subseteq [-a, a]^2$. For every sufficiently large $w > 0$, whenever $(k_1, k_2) \notin [-wa-1, wa]^2$, we have $\left\{[\frac{k_1}{w}, \frac{k_1+1}{w}] \times [\frac{k_2}{w}, \frac{k_2+1}{w}]\right\} \cap [-a, a]^2 = \emptyset$. This provides

$$\int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1, t_2)|\, dt_1 dt_2 = 0.$$

By Lemma 4.3, for every fixed $\epsilon > 0$, there exists a constant $M > 0$ (we can assume $M > a$) such that

$$\int_{\|\mathbf{x}\|_\infty > M} w^n |\chi(w\mathbf{x} - \mathbf{k})| d\mathbf{x} < \epsilon,$$

for every $(k_1, k_2) \in [-wa - 1, wa]^2$. For a set $\mathbf{G} = [-M, M]^2$. We compute on the complement of $\mathbf{G}$

$$\begin{aligned} J = &\int_{|x_2|>M} \phi_2 \left(\int_{|x_1|\leq M} \phi_1(\lambda|(K_w f)(x_1, x_2)|)\, dx_1 \right) dx_2 \\ &+ \int_{|x_2|\leq M} \phi_2 \left(\int_{|x_1|>M} \phi_1(\lambda|(K_w f)(x_1, x_2)|)\, dx_1 \right) dx_2 \end{aligned}$$

$$
\begin{aligned}
&+ \int_{|x_2|>M} \phi_2 \left(\int_{|x_1|>M} \phi_1(\lambda |(K_w f)(x_1, x_2)|)\, dx_1 \right) dx_2 \\
&:= J_1 + J_2 + J_3.
\end{aligned}
$$

First we compute J_1. In view of Jensen's inequality and Fubini-Tonelli theorem, we obtain

$$
\begin{aligned}
&\int_{|x_1|\le M} \phi_1 \left(\lambda \sum_{\frac{k_2}{w} \in [-a-\frac{1}{w}, a]} \sum_{\frac{k_1}{w} \in [-a-\frac{1}{w}, a]} |\chi(wx_1 - k_1, wx_2 - k_2)| w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1, t_2)| dt_1 dt_2 \right) dx_1 \\
&\le \frac{1}{m_0(\chi)} \sum_{\frac{k_2}{w} \in [-a-\frac{1}{w}, a]} \sum_{\frac{k_1}{w} \in [-a-\frac{1}{w}, a]} \phi_1 \left(\lambda m_0(\chi)\, w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1, t_2)| dt_1 dt_2 \right) \\
&\quad \times \int_{|x_1|\le M} |\chi(wx_1 - k_1, wx_2 - k_2)| dx_1 \\
&\le \frac{1}{m_0(\chi)} \sum_{\frac{k_2}{w} \in [-a-\frac{1}{w}, a]} \sum_{\frac{k_1}{w} \in [-a-\frac{1}{w}, a]} w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} \phi_1 \left(\lambda m_0(\chi) |f(t_1, t_2)| \right) dt_1 dt_2 \\
&\quad \times \int_{|x_1|\le M} |\chi(wx_1 - k_1, wx_2 - k_2)| dx_1 \\
&\le \frac{1}{m_0(\chi)} \sum_{\frac{k_2}{w} \in [-a-\frac{1}{w}, a]} w \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} \phi_1 \left(\lambda m_0(\chi) |f(t_1, t_2)| \right) dt_1 dt_2 \ \ \|\chi(\cdot, wx_2 - k_2)\|_{L^1([-M,M])}.
\end{aligned}
$$

Again using Jensen's inequality and Fubini-Tonelli theorem, we get

$$
\begin{aligned}
J_1 &\le \frac{1}{(m_0(\chi))^2} \sum_{\frac{k_2}{w} \in [-a-\frac{1}{w}, a]} \phi_2 \left(w \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} \phi_1 \left(\lambda \left(m_0(\chi) \right)^2 |f(t_1, t_2)| \right) dt_1 dt_2 \right) \\
&\quad \times \int_{|x_2|>M} \|\chi(\cdot, wx_2 - k_2)\|_{L^1([-M,M])} dx_2 \\
&\le \frac{1}{(m_0(\chi))^2} \sum_{\frac{k_2}{w} \in [-a-\frac{1}{w}, a]} \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \phi_2 \left(\int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} \phi_1 \left(\lambda \left(m_0(\chi) \right)^2 |f(t_1, t_2)| \right) dt_1 \right) dt_2 \\
&\quad \times \int_{|x_2|>M} w \|\chi(\cdot, wx_2 - k_2)\|_{L^1([-M,M])} dx_2 \\
&< \frac{\epsilon}{3 \left(m_0(\chi) \right)^2} I^{(\phi_1, \phi_2)} \left(\lambda \left(m_0(\chi) \right)^2 f \right).
\end{aligned}
$$

Note that the last inequality follows from the following observation:

$$
\int_{|x_2|>M} w \|\chi(\cdot, wx_2 - k_2)\|_{L^1([-M,M])}\, dx_2 < \frac{\epsilon}{3}.
$$

Similarly, we can estimate J_2. Thus we get

$$J_2 < \frac{\epsilon}{3\left(m_0(\chi)\right)^2} I^{(\phi_1,\phi_2)}\left(\lambda \left(m_0(\chi)\right)^2 f\right).$$

Let us rewrite J_3 as

$$J_3 := \int_{|x_2|>M} \phi_2\Bigg(\int_{|x_1|>M} \phi_1\Bigg(\lambda \sum_{\frac{k_2}{w}\in[-a,a]} \sum_{\frac{k_1}{w}\in[-a,a]} |\chi(wx_1-k_1, wx_2-k_2)| w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)| dt_1 dt_2 \Bigg) dx_1 \Bigg) dx_2.$$

In view of Jensen's inequality and Fubini-Tonelli theorem, we obtain

$$\begin{aligned}
&\int_{|x_1|>M} \phi_1 \left(\lambda \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \sum_{\frac{k_1}{w}\in[-a-\frac{1}{w},a]} |\chi(wx_1-k_1, wx_2-k_2)| w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)| dt_1 dt_2 \right) dx_1 \\
&\le \frac{1}{m_0(\chi)} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \sum_{\frac{k_1}{w}\in[-a-\frac{1}{w},a]} \phi_1\left(\lambda m_0(\chi)\, w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} |f(t_1,t_2)| dt_1 dt_2\right) \\
&\quad \times \int_{|x_1|>M} |\chi(wx_1-k_1, wx_2-k_2)| dx_1 \\
&\le \frac{1}{m_0(\chi)} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \sum_{\frac{k_1}{w}\in[-a-\frac{1}{w},a]} w^2 \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{\frac{k_1}{w}}^{\frac{k_1+1}{w}} \phi_1\left(\lambda m_0(\chi) |f(t_1,t_2)|\right) dt_1 dt_2 \\
&\quad \times \int_{|x_1|>M} |\chi(wx_1-k_1, wx_2-k_2)| dx_1 \\
&\le \frac{1}{m_0(\chi)} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} w \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} \phi_1\left(\lambda m_0(\chi) |f(t_1,t_2)|\right) dt_1 dt_2 \; \|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M]^c)}.
\end{aligned}$$

Again using Jensen's inequality and Fubini-Tonelli theorem, we have

$$\begin{aligned}
J_3 &\le \frac{1}{\left(m_0(\chi)\right)^2} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \phi_2\left(w \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} \phi_1\left(\lambda \left(m_0(\chi)\right)^2 |f(t_1,t_2)|\right) dt_1 dt_2 \right) \\
&\quad \times \int_{|x_2|>M} \|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M]^c)} dx_2 \\
&\le \frac{1}{\left(m_0(\chi)\right)^2} \sum_{\frac{k_2}{w}\in[-a-\frac{1}{w},a]} \int_{\frac{k_2}{w}}^{\frac{k_2+1}{w}} \phi_2\left(\int_{-a-\frac{1}{w}}^{a+\frac{1}{w}} \phi_1\left(\lambda \left(m_0(\chi)\right)^2 |f(t_1,t_2)|\right) dt_1 \right) dt_2 \\
&\quad \times \int_{|x_2|>M} w \|\chi(\cdot, wx_2-k_2)\|_{L^1([-M,M]^c)} dx_2.
\end{aligned}$$

Using Lemmma 4.3, we get

$$J_3 < \frac{\epsilon}{3\left(m_0(\chi)\right)^2} I^{(\phi_1,\phi_2)}\left(\lambda\left(m_0(\chi)\right)^2 f\right) .$$

On combining all the estimates, we deduce that

$$J < \frac{\epsilon}{\left(m_0(\chi)\right)^2} I^{(\phi_1,\phi_2)}\left(\lambda\left(m_0(\chi)\right)^2 f\right).$$

Step 3. For every $\epsilon > 0$, $\exists\, \delta > 0$ such that

$$\left(\int_{B_2}\int_{B_1} |\chi(u_1,u_2)| du_1 du_2\right) < \frac{\epsilon\left(m_0(\chi)\right)^2}{I^{(\phi_1,\phi_2)}(\lambda\left(m_0(\chi)\right)^2 f)} \tag{5.2}$$

for every pair of measurable sets $B_i \subset \mathbb{R}$ with $\mu(B_i) < \delta$, for $i = 1, 2$. Proceeding along the same lines as in the proof of Theorem 5.1 and absolute continuity property of the Lebesgue integral (5.2), we have

$$\begin{aligned}
&\int_{B_2} \phi_2\left(\int_{B_1} \phi_1(\lambda|(K_w f)(x_1,x_2)|)dx_1\right)dx_2\\
&\leq \frac{1}{\left(m_0(\chi)\right)^2} I^{(\phi_1,\phi_2)}(\lambda\left(m_0(\chi)\right)^2 f)\,\|\chi\|_{L^{(1,1)}(B_1\times B_2)}\\
&< \epsilon.
\end{aligned}$$

Thus, the integrals

$$\int_{B_2} \phi_2\left(\int_{B_1} \phi_1(\lambda|(K_w f)(x_1,x_2)|)dx_1\right)dx_2$$

are equi-absolutely continuous for arbitrary B_1 and B_2.

Therefore, by utilizing the Vitali convergence theorem along with the arbitrary choice of $\lambda > 0$, we obtain $I^{\overrightarrow{\Phi}}(\lambda(K_w f - f)) \to 0$ as $w \to \infty$. Similarly, we can extend the result for any $n \geq 3$. □

It is well established that $C_c(\mathbb{R})$ is dense in $L^{\phi}(\mathbb{R})$ [45, Theorem 7.6]. However, to the best of our knowledge, it has not been proven that $C_c(\mathbb{R}^n)$ is dense in $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$. To thoroughly explain this idea, we aim to provide detailed information as described below.

Lemma 5.4 *Let* $\overrightarrow{\Phi} \in \Delta_2$ *then the space* $C_c(\mathbb{R}^n)$ *is dense in* $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$.

Proof Our first step is to prove the result for $n = 2$. To establish the desired approximation, we demonstrate that any simple function can be closely approximated by a sequence of compactly supported continuous functions. Let $S \subset \mathbb{R}^2$ be a measurable set with $\mu(S) < \infty$. For each $m \in \mathbb{N}$, there exist an increasing sequence of compact set (U_m) and a decreasing sequence of open sets (V_m) such that $U_m \subset S \subset V_m$ and $\mu(V_m \setminus U_m) < 1/m$. By Urysohn's lemma, there exists $g_m \in C_c(\mathbb{R}^2)$ such that

$$g_m(\mathbf{x}) = \begin{cases} 1, & \text{for } \mathbf{x} \in U_m, \\ 0, & \text{for } \mathbf{x} \in \mathbb{R}^2 \setminus V_m. \end{cases}$$

It follows directly that for all $\mathbf{x} \in \mathbb{R}^2$, the function f_m satisfies

$$\mathbf{1}_{U_m}(\mathbf{x}) \leq g_m(\mathbf{x}) \leq \mathbf{1}_{V_m}(\mathbf{x}).$$

This implies that $0 \leq |g_m(\mathbf{x}) - \mathbf{1}_{U_m}(\mathbf{x})| \leq \mathbf{1}_{V_m \setminus U_m}(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^2$. For every $\gamma > 0$, we have

$$I^{\overrightarrow{\Phi}}(\gamma(\mathbf{1}_S - g_m)) \leq I^{\overrightarrow{\Phi}}(\gamma(\mathbf{1}_{V_m} - \mathbf{1}_{U_m})) \leq I^{\overrightarrow{\Phi}}(\gamma(\mathbf{1}_{V_m \setminus U_m})).$$

Since $|\mathbf{1}_{V_m \setminus U_m}(\mathbf{x})| \leq 1$ for all $\mathbf{x} \in \mathbb{R}^2$ and it converges to zero almost everywhere, the dominated convergence theorem for mixed-norm ([8, Section 2]) implies that $I^{\overrightarrow{\Phi}}(\gamma(\mathbf{1}_{V_m \setminus U_m}))$ approaches to *zero* as $m \to \infty$. This establishes that $I^{\overrightarrow{\Phi}}(\gamma(\mathbf{1}_S - g_m))$ approaches to *zero* as $m \to \infty$. In other words, for any given $\epsilon > 0$ and a simple function $h : \mathbb{R}^2 \to \mathbb{R}$, there exists a compactly supported continuous function $g : \mathbb{R}^2 \to \mathbb{R}$ such that

$$\|g - h\|_{\overrightarrow{\Phi}} < \frac{\epsilon}{2}. \tag{5.3}$$

It remains to prove that the class of simple functions is dense in $L^{\overrightarrow{\Phi}}(\mathbb{R}^2)$. For $f \in L^{\overrightarrow{\Phi}}(\mathbb{R}^2)$, there exists an increasing sequence of simple functions $(f_n)_{n\in\mathbb{N}}$ such that $0 \leq f_n \uparrow |f|$ pointwise as $n \to \infty$. Moreover, for any $\alpha > 0$, we have $\phi_1(\alpha(|f| - f_n)) \leq \phi_1(\alpha f)$ and $\phi_1(\alpha(|f| - f_n))$ approaches to zero as $n \to \infty$. Hence, by the *dominated convergence theorem* for mixed-norm, we have

$$\lim_{n\to\infty} \int_{\mathbb{R}} \phi_2 \left(\int_{\mathbb{R}} \phi_1 \left(\alpha |(|f| - f_n)(x_1, x_2)|\right) dx_1 \right) dx_2 = 0.$$

Hence, for any positive function $|f| \in L^{\overrightarrow{\Phi}}(\mathbb{R}^2)$ there exists a non-negative simple function arbitrary close to $|f|$.

As $f \in L^{\overrightarrow{\Phi}}(\mathbb{R}^2)$, then we can write $f = f^+ - f^-$, where $f^+, f^- \in L^{\overrightarrow{\Phi}}(\mathbb{R}^2)$ are non-negative functions defined as follows:

$$f^+(x) = \max\{f(x), 0\} \quad \text{and} \quad f^-(x) = -\min\{f(x), 0\}.$$

Hence, for any $\epsilon > 0$ there exist non-negative simple functions s^+ and s^- such that

$$\|f^+ - s^+\|_{\overrightarrow{\Phi}} < \frac{\epsilon}{4}, \quad \|f^- - s^-\|_{\overrightarrow{\Phi}} < \frac{\epsilon}{4}.$$

Therefore for any $f \in L^{\overrightarrow{\Phi}}(\mathbb{R}^2)$ and $\epsilon > 0$, there exists a simple function $s := s^+ - s^-$ such that

$$\|f - s\|_{\overrightarrow{\Phi}} \leq \|f^+ - s^+\|_{\overrightarrow{\Phi}} + \|f^- - s^-\|_{\overrightarrow{\Phi}} < \frac{\epsilon}{2}. \tag{5.4}$$

In view of the triangle inequality of mixed-norm Orlicz space, and the equations (5.3) and (5.4), we get $\|f - h\|_{\overrightarrow{\Phi}} < \epsilon$.

This establishes the result for $n = 2$. Following the similar lines of proof, one can extend this proof for any $n \geq 3$. □

We now proceed to prove the main theorem of this section, namely the convergence of (4.1) in mixed-norm Orlicz space $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$.

Theorem 5.5 *Let $f \in L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$ be fixed and $\overrightarrow{\Phi} \in \Delta_2$. Then we have*

$$\lim_{w\to\infty} \|K_w f - f\|_{\overrightarrow{\Phi}} = 0.$$

Proof We prove the result using the denseness of $C_c(\mathbb{R}^n)$ in $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$ as established in Lemma 5.4. Let $f \in L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$ and $\epsilon > 0$ be fixed. Thus, there exists $g \in C_c(\mathbb{R}^n)$ such that

$$\|f - g\|_{\overrightarrow{\Phi}} < \frac{\epsilon}{2\left(1 + \left(m_0(\chi)\right)^n\right)}. \tag{5.5}$$

Again Theorem 5.3 implies there exists $N_0 \in \mathbb{N}$ such that for each $w \geq N_0$, we have

$$\|K_w g - g\|_{\overrightarrow{\Phi}} < \frac{\epsilon}{2}.$$

Now for $w \geq N_0$, the triangle inequality of $\|\cdot\|_{\overrightarrow{\Phi}}$ along with (5.5) and Remark 5.2, we obtain

$$\begin{aligned}
\|K_w f - f\|_{\overrightarrow{\Phi}} &\leq \|g - f\|_{\overrightarrow{\Phi}} + \|K_w f - K_w g\|_{\overrightarrow{\Phi}} + \|K_w g - g\|_{\overrightarrow{\Phi}} \\
&< \left(1 + \left(m_0(\chi)\right)^n\right) \|g - f\|_{\overrightarrow{\Phi}} + \frac{\epsilon}{2} \\
&< \epsilon.
\end{aligned}$$

This completes the proof. □

6 Examples of kernel

Kernels hold significant importance in the theoretical framework. In this section, we present several examples of suitable kernels that satisfy assumptions (3.1) and (3.2) of our presented theory. Assume that for each $i = 1, \ldots, n$, $\chi_i : \mathbb{R} \to \mathbb{R}$ is a kernel satisfies

$$\sum_{k_i\in\mathbb{Z}} \chi_i(u_i - k_i) = 1, \qquad u_i \in \mathbb{R},$$

and

$$m_0(\chi_i) := \sup_{u_i \in \mathbb{R}} \sum_{k_i \in \mathbb{Z}} |\chi_i(u_i - k_i)| < +\infty.$$

Then the kernel $\chi : \mathbb{R}^n \to \mathbb{R}$ is defined by

$$\chi(\mathbf{u}) = \prod_{i=1}^{n} \chi_i(u_i),$$

belongs to $L^1(\mathbb{R}^n)$, as

$$\int_{\mathbb{R}^n} |\chi(\mathbf{u})| d\mathbf{u} = \int_{\mathbb{R}^n} \prod_{i=1}^{n} |\chi_i(u_i)| du_1 ... du_n = \prod_{i=1}^{n} \int_{\mathbb{R}} |\chi_i(u_i)| du_i < \infty.$$

Moreover, from the above observations, we have

$$\sum_{\mathbf{k} \in \mathbb{Z}^n} \chi(\mathbf{u} - \mathbf{k}) = \prod_{i=1}^{n} \sum_{k_i \in \mathbb{Z}} \chi_i(u_i - k_i) = 1,$$

and

$$m_0(\chi) = \sup_{\mathbf{u} \in \mathbb{R}^n} \sum_{\mathbf{k} \in \mathbb{Z}^n} |\chi(\mathbf{u} - \mathbf{k})| = \prod_{i=1}^{n} m_0(\chi_i) < +\infty.$$

Remark 6.1 *([6])* (a) If $\chi(x) = \mathcal{O}(x^{-1-\epsilon-\alpha})$ for $x \to \pm\infty$ and some $\epsilon > 0$, the kernel χ satisfies (3.2).
(b) The condition (3.1) is equivalent to the following condition

$$\widehat{\chi}(k) = \begin{cases} 1, & k = 0 \\ 0, & k \neq 0, \end{cases}$$

where $\widehat{\chi}(u) := \int_{\mathbb{R}} \chi(v) e^{-ivu} \, dv$, $u \in \mathbb{R}$ is the Fourier transform of χ.

Next, we provide several important examples of kernels that satisfy the assumptions of our proposed theory.

(I) Fejér kernel: The Fejér kernel is defined as ([6, 25])

$$F(x) := \frac{1}{2} \operatorname{sinc}^2\left(\frac{x}{2}\right), \qquad x \in \mathbb{R}.$$

It can be observed that the Fourier transform of sinc-function is given by

$$\widehat{\operatorname{sinc}}(v) = \begin{cases} 1 & |v| \leq \pi, \\ 0, & |v| > \pi. \end{cases}$$

Indeed $F \in L^1(\mathbb{R})$ is bounded. In view of Remark 6.1(a), one can verify that F satisfies (3.2). Furthermore, the Fourier transform of F is given by

$$\widehat{F}(v) = \begin{cases} 1 - |\frac{v}{\pi}|, & v \leq \pi, \\ 0, & v > \pi. \end{cases}.$$

Let $\mathcal{F}(\mathbf{x}) := \prod_{i=1}^{n} F(x_i)$ be the multivariate Fejér's kernel. Then $\mathcal{F}$ fulfills the conditions (3.1) and (3.2) associated with the generalized kernel (see [25]).

In order to reduce truncation error, one can take $\chi(x)$ with faster decay as $x \to \pm\infty$. One prominent example in this direction is the Jackson kernel.

(II) Jackson kernel: The Jackson kernel with parameters $\beta \geq 1, m \in \mathbb{N}$ is given by ([6])

$$\overline{J_{\beta,m}}(x) := C_{\beta,m}\ \mathrm{sinc}^{2m}\left(\frac{x}{2\beta m\pi}\right) \quad (x \in \mathbb{R}),$$

with $C_{\beta,m} := \left[\int_{\mathbb{R}} \mathrm{sinc}^{2m}\left(\frac{x}{2\beta m\pi}\right) dx\right]^{-1}$.

It can be observed that the Fourier transform of the Jackson kernel vanishes outside the interval, i.e., $\overline{J_{\beta,m}}(x)$ are bandlimited to $[-1/\beta, 1/\beta]$. As a consequence of Remark 6.1(a) and (b), we conclude that (3.1) and (3.2) are satisfied.

As in the previous case, the multivariate Jackson kernel given by $\mathcal{J}(\mathbf{x}) = \prod_{i=1}^{n} \overline{J_{\beta,m}}(x_i)$, satisfies all the desired properties.

One can minimize the truncation error by selecting a suitable kernel χ with compact support. A noteworthy example in this direction is the well-known B-spline kernel.

(III) B-spline kernel: The B-spline kernel of order $m \in \mathbb{N}$ is defined as ([6, 25])

$$B_m(t) := \frac{1}{(m-1)!}\sum_{j=0}^{m}(-1)^j\binom{m}{j}\left(\frac{m}{2}+t-j\right)_+^{m-1}, \qquad t \in \mathbb{R},$$

where $(x)_+ := \max\{x, 0\}$, $x \in \mathbb{R}$. The Fourier transform of B_m is given by

$$\widehat{B_m}(v) = \mathrm{sinc}^m\left(\frac{v}{2\pi}\right).$$

Based on Remark 6.1(b) we conclude that B_m satisfies (3.1). Furthermore, B_m is bounded on $\mathbb{R}$ for all $m \in \mathbb{N}$, with compact support contained in $[-m/2, m/2]$. Therefore, we can say that $B_m \in L^1(\mathbb{R})$. Moreover, the moment condition is satisfied for all $\alpha > 0$. As in the previous case, the multivariate B-spline kernel can be written as

$$\mathcal{B}(\mathbf{x}) = \prod_{i=1}^{n} B_m(x_i),$$

which satisfies all the properties of a kernel.

Another example in this direction is the Bochner–Riesz kernel.
(IV) Bochner-Riesz kernel: The Bochner–Riesz kernel is defined as ([13])

$$B^{\gamma}(\mathbf{x}) = \frac{2^{\gamma}}{\left(\sqrt{2\pi}\right)^{N}} \Gamma(\gamma+1)\,(|\mathbf{x}|)^{-(N/2-\gamma)}\, J_{(N/2)+\gamma}(|\mathbf{x}|), \quad \mathbf{x} \in \mathbb{R}^n$$

for $\gamma > 0$, where J_{λ} is the Bessel function of order λ and Γ is Gamma function. The Fourier transform of B^{γ} is given by

$$\widehat{B}^{\gamma}(\boldsymbol{\xi}) = \begin{cases} (1-|\boldsymbol{\xi}|^2)^{\gamma}, & |\boldsymbol{\xi}| \leq 1, \\ 0, & |\boldsymbol{\xi}| > 1. \end{cases}$$

In view of Remark 6.1, we conclude that B^{γ} satisfies (3.1) and (3.2).

7 Conclusion and future work

7.1 Conclusion

This article aims to study the approximation properties of a well-known family of Kantorovich type sampling operators in a broad framework of mixed-norm function spaces. This not only contributes to approximation theory, but also explores mixed-norm spaces in function theory. To analyze the boundedness of generalized sampling operators in mixed-norm Lebesgue spaces, we have introduced an appropriate subspace of $L^{\overrightarrow{P}}(\mathbb{R}^n)$, denoted by $\Delta^{\overrightarrow{P}}(\mathbb{R}^n)$, in Sect. 3. Further, we have established the boundedness and convergence of Kantorovich-type sampling operators within the broad framework of mixed-norm Lebesgue spaces and mixed-norm Orlicz spaces. In order to establish the proposed results, we have proved a fundamental density result stating that the space of compactly supported continuous functions $C_c(\mathbb{R}^n)$ is dense in mixed-norm Orlicz space $L^{\overrightarrow{\Phi}}(\mathbb{R}^n)$.

7.2 Limitations

The limitations of the present article are summarized as follows.

- The results obtained in the mixed-norm Lebesgue space are established under suitable restrictions on the exponents p_i, such as the monotonicity condition $p_1 \leq p_2 \leq \ldots \leq p_n$, and therefore may not cover mixed-norm setting completely.
- The convergence analysis of the generalized sampling operator is carried out within a suitable subspace of mixed-norm Lebesgue spaces, where the sampling series is well-defined; consequently, the results do not automatically extend to the whole ambient space.
- In the mixed-norm Orlicz spaces, some results require the assumption that the Orlicz function must satisfy Δ_2-condition mentioned in Definition 2.4 . Hence, the corresponding results may not cover the most general class of Orlicz spaces.

7.3 Future direction

A natural direction for future research is to establish the order of convergence and quantitative approximation results for generalized and Kantorovich-type sampling operators by employing suitable notion of modulus of continuity. Alongwith with these theoretical developments, we intend to investigate the applicability of the proposed setting in the real-world image processing tasks, such as image reconstruction, image inpainting, and image denoising using realistic data sets as it provides a more flexible and effective tool for modeling anisotropic features.

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Data Availability Data sharing is not applicable to this article, as no data sets were generated or analyzed during the current study.

Declarations

Conflicts of Interest The author declares no conflict of interest related to the content of this article.

Ethic The authors declare that the present work is original and has not been published elsewhere nor is it under consideration for publication elsewhere. No part of the manuscript has been copied or plagiarized from other sources, and all results and references have been properly acknowledged.

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